Thermodynamic description of the Ag–Cu system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Ag–Cu binary system as calculated with pycalphad from the SGTE dataset AgCu-06He.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, L → (Ag) + (Cu), lies at 1055.8 K.

1. Introduction

The Ag–Cu system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as simple eutectic. The thermodynamic description is the SGTE collection entry [1], which reproduces the assessment of X.C. He, H. Wang, H.S. Liu, Z.P. Jin, Calphad, 30, 367-74(2006). [2]. The calculations use pycalphad 0.11.2 at 1 atm.

Pure Ag melts at 1234.9 K. The melting point of Cu reproduced by the dataset is 1357.8 K.

2. Phase diagram and invariant reactions

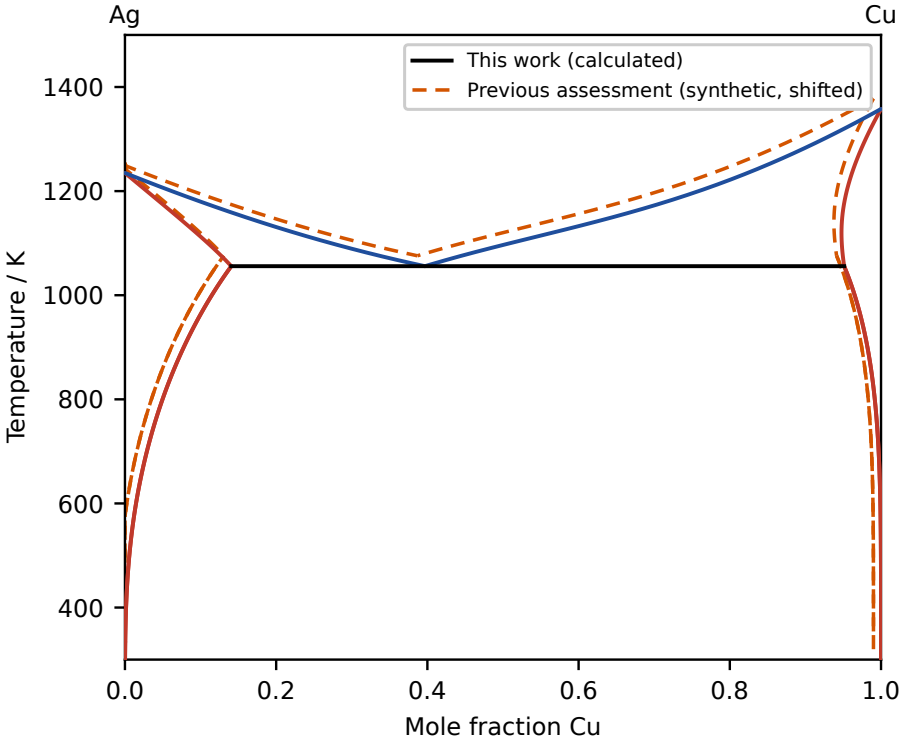


Fig. 1. Calculated Ag–Cu phase diagram and a dashed, deliberately shifted second assessment.

The dataset places a eutectic (L → (Ag) + (Cu)) at 1055.8 K; the composition of L is 39.7 at.%.

Table 1. Invariant reactions and congruent points, calculated (Ag–Cu).

Reaction	Type	T (K)	x(Cu), 1	x(Cu), 2	x(Cu), 3
L → (Ag) + (Cu)	eutectic	1055.8	0.397	0.141	0.951

Phases 1–3 in the order written in the reaction. Calculated with pycalphad 0.11.2 from the SGTE dataset; synthetic document.

3. Phase stability

Table 2. Solid phases: structure, stability range and homogeneity range (Ag–Cu).

		Structure		Stability range / °C		Homogeneity range	
Phase	Pearson	Strukturb.	Prototype	from	to	at.% Cu	at T / °C
(Ag)	cF4	A1	Cu	< 27	961.8	0.0–13.8	787
(Cu)	cF4	A1	Cu	< 27	1084.6	94.8–100.0	847

According to Table 2, (Ag) remains stable up to 1234.9 K (melting of the pure element). The solubility of Ag in (Cu) reaches 5.2 at.% at 1120.0 K.

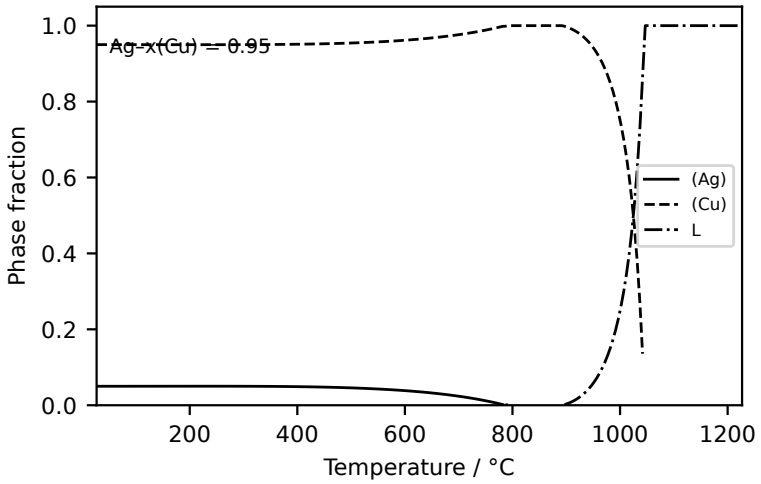


Fig. 2. Calculated equilibrium phase fractions versus temperature for Ag–x(Cu) = 0.95.

4. Thermodynamic properties

At 1450 K the integral mixing enthalpy of the liquid has its maximum, 3683 J/mol, at 54.0 at.% Cu. For the equiatomic liquid at 1450 K, the Gibbs energy of mixing is −5394 J/mol.

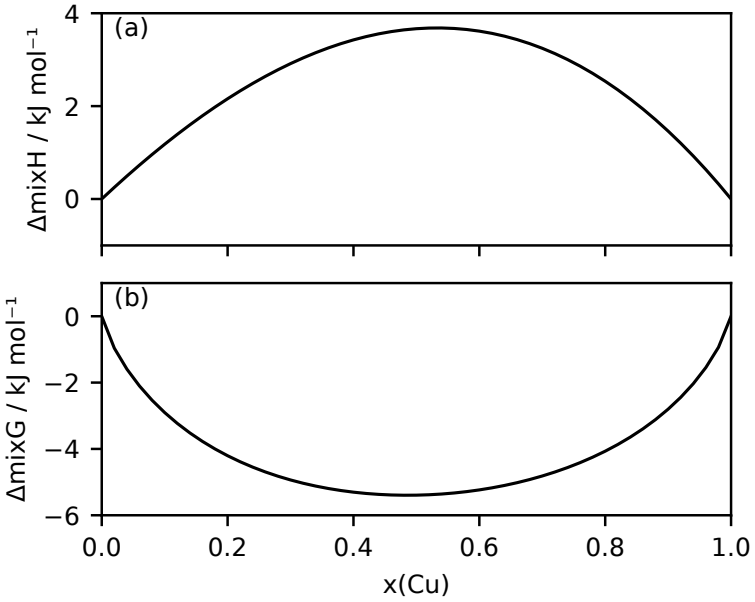


Fig. 3. Calculated (a) mixing enthalpy and (b) mixing Gibbs energy of the liquid at 1450 K.

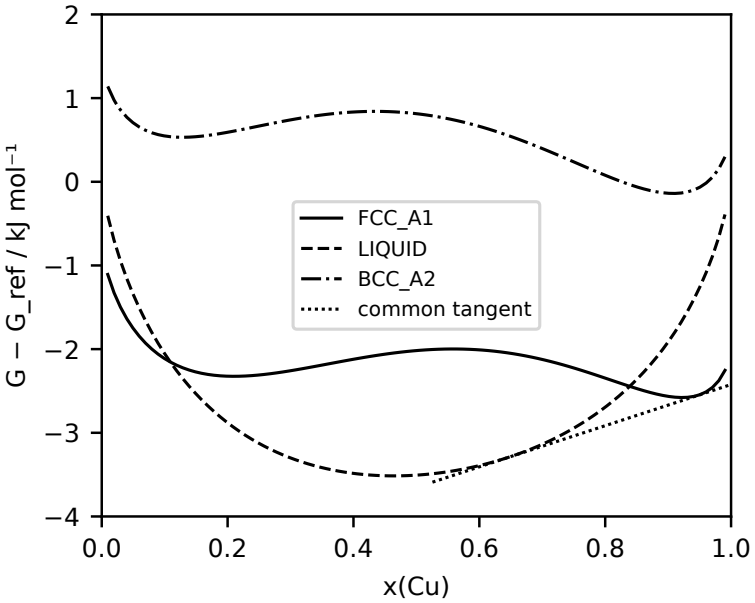


Fig. 4. Calculated Gibbs energy curves at 1150 K.

Table 3. Redlich–Kister parameters of the liquid (Ag–Cu).

ν	a / J mol⁻¹	b / J mol⁻¹ K⁻¹	c / J mol⁻¹ K⁻¹
L0	16914.949	-14.7721	1.54955
L1	-1963.3	0.8623	0.0

Table 4. Integral and partial mixing properties of the liquid at 1450 K (Ag–Cu).

	Liquid at 1450 K					
x(Cu)	ΔmixH / kJ mol⁻¹	ΔmixG / kJ mol⁻¹	a(Ag)	a(Cu)	ΔH(Ag) / kJ mol⁻¹	ΔH(Cu) / kJ mol⁻¹
0.100	1.18	-2.90	0.9075	0.2154	0.10	10.93
0.200	2.16	-4.21	0.8278	0.3724	0.41	9.14
0.300	2.92	-4.94	0.7575	0.4885	1.00	7.38
0.400	3.43	-5.30	0.6930	0.5772	1.91	5.70
0.500	3.67	-5.39	0.6299	0.6488	3.18	4.16
0.600	3.61	-5.24	0.5626	0.7116	4.86	2.79
0.700	3.25	-4.82	0.4828	0.7721	6.99	1.64
0.800	2.54	-4.07	0.3780	0.8364	9.64	0.76
0.900	1.46	-2.80	0.2282	0.9103	12.84	0.20

Reference states: pure liquid Ag and Cu at 1450 K; per mole of atoms. ΔH: partial molar enthalpies of mixing.

5. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.

References

[1] B. Hallstedt, SGTE binary dataset collection, Calphad 89 (2025) 102833 (CC BY 4.0).

[2] X.C. He, H. Wang, H.S. Liu, Z.P. Jin, Calphad, 30, 367-74(2006).

[3] K. Synthetica, L. Exemplar, J. Fict. Phase Equilib. 12 (1987) 345–352 — fictitious reference, invented for this synthetic benchmark.

[4] M. Placeholder, Z. Notreal, Fict. Thermochim. Acta 41 (1994) 77–89 — fictitious reference, invented for this synthetic benchmark.

[5] R. Imaginary, Proc. Synth. Calorim. Soc. 8 (2003) 1102–1110 — fictitious reference, invented for this synthetic benchmark.